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ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА

Волнения и оппозиционные СМИ при автократии: свидетельства из царской России

Программа Бакалавр экономики

Совместная программа по экономике НИУ ВШЭ и РЭШ

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Unrest and Opposition Media in Autocracy: Evidence from Tsarist Russia

Аннотация

Последние 20 лет существования Российской империи характеризовались социальным и политическим внутренними кризисами. Радикальные левые движения набирали все большую популярность, а крестьянские восстания и стачки рабочих значительно участились. В своей выпускной работе я анализирую взаимосвязь между радикальными политически левыми изданиями и народными волнениями того периода. Используя гравитационную модель, я обнаружил статистически и экономически значимую положительную корреляцию между стачками рабочих и левой прессой, в то время как для крестьянских восстаний такая корреляция отсутствовала. Также мной были найдены доказательства в пользу гипотезы о том, что наличие корреляции в случае рабочих было обусловлено их более высоким уровнем грамотности по сравнению с крестьянами. Таким образом, пропаганда левых партий либо не оказывала на неграмотных крестьян эффекта, либо, мысля стратегически, левые партии могли не распространяли свою периодику среди крестьян вовсе.

Abstract

The last 20 years of the Russian Empire were characterized by social and political internal crises. Radical leftist movements were gaining popularity, while peasant uprisings and worker strikes were becoming much more frequent. In my thesis, I analyze the relationship between radical political leftist periodicals and the popular unrest of the period. Using gravity model, I found a statistically and economically significant positive correlation between worker strikes and the leftist press, while there was no such correlation for peasant revolts. I also found evidence in favor of the hypothesis that the presence of correlation in the case of workers was due to their higher literacy rate compared to peasants. Thus, the propaganda of the leftist parties either had no effect on the illiterate peasants or, thinking strategically, the leftist parties did not distribute their periodicals among peasants at all.

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1 Introduction

At the beginning of the 20th century the Russian Empire was in a state of social turmoil. In its last years, the country experienced three revolutions; peasant uprisings and worker strikes became much more frequent. Left-wing party agitators took an active part in these events by distributing their periodicals. The agitators got more leeway after 1905, when censorship was relaxed during the First Russian Revolution and left-wing parties were able to legally print their publications. The question arises here: what was the association between the popular unrest and the leftist periodicals? It is important to emphasize that the issue will be examined within the framework of an authoritarian semi-agrarian state of the early 20th century. The regime of the Russian Empire can indeed be described as tsarist autocracy (samoderzhaviye), in which the entirety of actual power belonged to the emperor. The emperor appointed the highest officials, commanded the army, controlled all three branches of government and had the exclusive right to pardon. I also consider the Russian Empire a semi-agrarian country, as the main economic sector at that time was still agriculture, although the country had embarked on the path of industrialization of its economy. In my opinion the study of the correlation between left media and popular unrest of the time is important, since it will help researchers better understand the social processes taking place not only in the Russian Empire, but also in other authoritarian semi-agrarian countries of the time: Bulgaria, Serbia, Romania, Portugal, etc. Moreover, this question is of interest within the framework of media economics as well. The major motivation for my research is the new autocratic setting. Many modern media economic studies are conducted on the examples of democratic states, such as the United States, due to the greater availability of data. Thus, with my research, I will try to expand the scope of media economic analysis to include autocratic regimes and their unique challenges. In addition, the object of my research is traditional media, particularly newspapers, so my results may be useful for analyzing the media of contemporary impoverished countries with low internet and telephone coverage, where newspapers may still be a key source of information. In my work, I examine this correlation for two types of popular discontent: peasant revolts and worker strikes. As a leftist press, I will consider publications affiliated with labor parties: Russian Social Democratic Labor Party (RSDLP), Socialist Revolutionary Party (SRs), Trudoviks, Anarchists, etc. The Constitutional Democratic Party, for example, though moderately leftist, were not a labor party, so I don't consider their press leftist. It should be noted that not all the newspapers in my source had a clear party affiliation: a large proportion of them were labeled as either "bourgeois" publications or "labor" publications. Therefore, I do the analysis specifically for aggregate press metrics of all left-wing labor parties together to avoid biased estimations.

2 Literature review

The main focus of my research is to study the correlation of the politically leftist press in the gubernias of the Russian Empire with the number of peasant revolts and worker strikes. To begin with, a few articles on media economics should be mentioned to understand how media can influence its audience. In *Political Talk Radio and Public Opinion* (Barker & Knight, 2000) the researchers found that listening to political Limbaugh radio talk show had a significant effect on listeners' political views. The magnitude of the effect was directly related to the amount of time devoted in the radio broadcast to a particular topic. Interestingly, radio was much more effective in creating opposition than in building support for particular views. Thus, people were more willing to listen to negative statements about groups, people and ideas than to positive ones. This reinforces the motivation for my research, for if the mechanisms of media influence on public consciousness are indeed arranged in this way, then leftist labor parties openly criticizing the tsarist regime may have been successful in inciting popular discontent. In addition, in *Surprise me if you can: The influence of newspaper endorsements in U.S. presidential elections* (Casas & Fawaz, 2016) the authors of the article evaluated the effect of endorsement of presidential candidates in 2008 and 2012 by print newspapers on public opinion. It was shown that print media had a significant effect on readers' opinion. The effect was greater when the candidate endorsed by the newspaper was both consistent and unexpected. In other words, a candidate received the most promotion if his or her figure did not contradict the political views of the newspaper' audience, while the candidate's party itself had not been previously endorsed by the newspaper. The authors used prediction-market data (the pricing of future contracts on a candidate's victory) to analyze the probabilities of candidates winning. As a part of their identification strategy, they made an assumption that endorsements of newspapers and their timing didn't depend on the candidate's winning probability. Thus, they aren't concerned with the endogeneity problem, saying that they have daily observations. Unfortunately, this approach does not work for my research, as I operate with annual data. Considering the topic of media economics, the paper *Media Bias and Influence: Evidence from Newspaper Endorsements* (Chiang and Knight, 2011) is important for my research as well. The researches found similar effect that voters rely on the media during election campaigns but that the extent of this reliance depends on the credibility of the endorsement. Moreover, the group most heavily exposed to the media were moderate voters with no strict political stance. According to the researchers, people understand the political bias of periodicals and tend to discount the information they receive on it. Thus, endorsement of a Democratic candidate from a left-wing newspaper will be less credible than endorsement of a Republican candidate from the same newspaper. In my data, there is no such precise differentiation of newspapers by partisanship and endorsement of candidates from other parties. Because of

this limitation, I will have to neglect source credibility as an element of my analysis. To address the problem of endogeneity, researchers use "the fraction of readers supporting the Democrat prior to the publication of the endorsement." [1] This approach would also be a poor fit for my study due to the lack of data on newspaper audience support for a particular party, as well as the use of panel data with annual observations. However, together with the paper of Casas & Fawaz researchers prove that traditional media, like newspapers, do have an impact on behavior of people even these days, let alone in the early 20th century when it was practically the only media resource available, enhancing my motivation. Also, it is important to note that both of these studies were conducted within the framework of a democratic state (the USA). Thus, my study will well complement the findings of these authors, considering the fact that Russian Empire was an autocratic state. Similarly, Enikolopov, Petrova, and Zhuravskaya in *Media and Political Persuasion: Evidence from Russia* (2010) showed that the access of audience to the independent media (TV channels) decreased the aggregate vote for the government party in Russian 1999 Parliamentary Election. As in previous researches, the degree of the influence depended on the credibility of the media source. Complementing these studies, the researcher in *News and Public Opinion: Which Comes First?* (Wlezein, 2023) showed the inverse effect of audience on media. The author concluded that public opinion influences news covers. Reverse effect is also present, but the media effects were less frequent and relatively smaller than those that flow from the public to the news. The magnitude of both the effect of the media on the behavior of its audience and the audience on news covers is directly related to whether new information about the agent was disclosed. For example, news with exclusive political information had a comparatively greater impact on the audience. Thus, from these studies it can be concluded that media can indeed influence the actions of its audience, but there may be a reverse effect where the audience shapes media trends. This is why the potential finding of a statistically significant correlation between popular unrest and the left-wing press should not be taken as a causal relationship, because the effect could be in the other direction.

It is crucial for my research to consider the literature related to popular discontent as well. There are several scientific researches directly touching upon the problematic of the causes of popular discontent in the Russian Empire of the early 20th century. Among them it is necessary to highlight the study *Career incentives in political hierarchy: evidence from Imperial Russia* (Gokmen and Kofanov, 2020). One of the main results obtained by the authors is that there is no effect of career incentives on governors' performance in the gubernias. A statistically significant effect was found only for oblasts: regions which were distant from the center and poorly developed. Thus, the state had no mechanism of

[1] Chiang, C.-F. and B. Knight (2011). Media bias and influence: Evidence from newspaper endorsements. The Review of economic studies 78 (3), 795–820.

rewards and punishments, which could lead to mismanagement of governors over their gubernias manifested in popular unrest. I have no data on the personal qualities of governors, but the differences in the levels of their responsibility in my regressions implicitly will be accounted for using gubernia and year fixed effects as well as linear gubernia trends. The data of researchers on peasant revolts also is used in my work. One more extremely important work in this field is the article *(Good) Land and Freedom (for Former Serfs): Determinants of Peasant Unrest in European Russia, March-October 1917* (Finkel, Gehlbach, and Kofanov, 2017). The authors of this study tried to identify the main determinants of peasant revolts in the period from March to October 1917, concluding that the quality of the soil and the historical prevalence of serfdom in the region had the great impact on the popular uprisings. In their study, they did not analyze political determinants, like the correlation between media resources and popular unrest, so my research may add to their findings. This research also underlines the importance of using fixed effects in my regressions, to clear the variation in popular discontent from historically and geographically determined causes. In addition, it's worth mentioning the study *Collective Action and Representation in Autocracies: Evidence from Russia's Great Reforms* (Castaneda Dower, Finkel, Gehlbach, and Nafziger, 2017), in which the authors analyzed the consistency of the results of the Acemoglu-Robinson model on the example of peasant uprisings in the Russian Empire under Alexander 2. The researchers found that peasants had relatively lower representation in regions with a high historical probability of revolts. Presumably, this may have been because the authorities either wanted to punish peasants for disobedience or were unwilling to conduct liberalization in dangerous regions, fearing only to give the rioters more opportunities. The results of the study *Famine, Inequality, and Conflict* (Meriläinen & Mitrunen, 2022) are also important for my paper. The researchers showed that the 1866-1868 famine in Finland was a significant determinant of the redistribution of wealth to large landowners. This led to the regions most affected by the famine being the most actively involved in the Finnish Civil War. Complementing the results of this study, I should mention the paper *"Let Them Eat Cake": Drought, Peasant Uprisings, and Demand for Institutional Change in the French Revolution* (Waldinger, 2023). This study demonstrated that peasants in areas of France most severely affected by drought in 1788 exhibited comparatively higher levels of popular discontent (rebellion) during the Great French Revolution of 1789. It was found that the demand for change in social and economic institutions was comparatively higher in the most drought-affected areas. This was measured through complaints filed by peasants. Thus, the level of agricultural yields have a significant impact on violent uprisings. These researches on French and Finnish data emphasize the importance of controlling for yield in my specifications. In my paper, I will control for the productivity of the agricultural sector and worker wages, which reflect people's standard of living, as one of the most important time-varying variables that can affect popular discontent.

3 Data

In my work I analyze 49 gubernias located in European part of Russian Empire. I use data on periodicals from 1901 to 1913 using the reference book by *Cherepakhov and Ingerit* (1957), namely information on the number of unique left-wing periodicals in the gubernia per year, the dates of the beginning and end of the publication, the party affiliation of the publication and its circulation. It's important to note that these data are new and they were classified and digitised for the very first time by me manually. A periodical was considered left-wing if it was affiliated with labor parties such as the RSDLP, Left SRs, Trudoviks, Anarchists, etc. As it was already mentioned, moderately leftist non labor parties, like the Constitutional Democratic Party, aren't considered leftist in my work. The source did not always accurately indicate the newspaper's affiliation with a particular party. Often researchers noted that periodical was "bourgeois", "labor", or controlled by the "tsarist government". In such cases, I considered only "labor" periodicals. It is worth nothing that this source was compiled in the USSR, which creates grounds for political bias. The Soviet government could influence the scientists gathering this data not to include some periodicals in the collection. I assume that Soviet scholars tried to document leftist labor parties periodicals in the most complete way possible (which is partially confirmed by the presence in the source of a large amount of illegal leftist press printed on hectographs). Hence, when studying the correlation between the leftist labor press and popular discontent, the under-registration issue is unlikely.

The main dependent variables in my study are peasant revolts and worker strikes. The data on peasant revolts from 1901 to 1913 were taken from the database of Dmitriy Kofanov used in the research *Career incentives in political hierarchy: evidence from Imperial Russia* (Gokmen & Kofanov, 2020). I took data on number of workers, worker strikes, the number of workers participated in strikes, and workers wages from the Reports of factory inspectors. All these data were also collected, classified and digitised manually by me from the scans of the reports of factory inspections. It is important to note that the inspectors' reports do not include data on the strikes of 1905-1908, since during the First Russian Revolution it was difficult to count strikes and number of workers participated in them due to the mass scale nature of those strikes. Thus, in this research I used the data on worker strikes from 1901-1904 and 1909-1913. Also, in the calculation of number of workers per gubernia the inspection took into account only the workers who were imposed to pay fines. Thus, the parameter of average wage per worker, obtained by dividing the total of all workers' incomes by their number, is noisy. This is a limitation of my dataset that could potentially bias the estimate of the correlation of interest, but there is no better quality data for worker wages.

$$\text{Wage}_{it} = \frac{\text{Total earnings}_{it}}{\text{Number of workers}_{it}}$$

I also use agricultural productivity in the gubernias in my work as a control variable, following the bachelor thesis *The Effect of Economic Shocks on Peasant Uprisings: Evidence from the Russian Empire* by Ilya Gilyov (2021). Data on the amount of crops sown and harvested up to 1900 were taken from the article *The economic effects of the abolition of serfdom: Evidence from the Russian empire* (Markevich and Zhuravskaya, 2018). I used this data from 1890-1899 to calculate yield (the number of poods harvested divided by the poods sown) constant trend for each gubernia.

$$\text{Productivity trend}_i = \frac{\frac{\text{harvest}_{1890}}{\text{seeds}_{1890}} + \ldots + \frac{\text{harvest}_{1899}}{\text{seeds}_{1899}}}{10}, \quad i - \text{gubernia identifier}$$

Data on yield from 1900 to 1913 are taken from the *New Russian economic history* (Guriev, Markevich, and Zhuravskaya, 2021). Thus, I construct the agricultural productivity parameter as follows:

$$\text{Productivity}_{it} = \frac{\text{harvest}_{it}}{\text{seeds}_{it}} - \text{Productivity trend}_i$$

A also used data on the number of printing houses for 1904 in each gubernia from the master's thesis of Andrey Kolyadenko *How Media Contributes to Unrest in Autocracy: a Case of Imperial Russia* (2016). In addition, I used data from the 1897 census of the Russian Empire for urban and rural populations with their division into literate and illiterate.

I will analyze the peasant revolts in the 1901-1913 time period, with drop of 1905 and 1906 years. I removed these years from the sample because it was the active phase of the First Russian Revolution. I consider these years as outliers, since the sharp increase in peasant revolts and worker strikes during the revolution was due to complex economic and political reasons: failures in the Russo-Japanese War, the shooting of peaceful demonstrators by the imperial guard during Bloody Sunday, the lack of political representation for the majority of the population, harsh working conditions in the industrial sector. Therefore, I believe that it would be misleading to study the correlation between the leftist press and popular unrest in these years, because they are not representative and the correlation would include effects from other uncontrolled sources. For workers strike I will analyze 1901-1904 and 1909-1913 time interval, due to the absence of data on strikes for 1905-1908.

	Peasant revolts: Summary statistics					
	sum	mean	sd	min	max	count
Revolts rural	1.638904	.0030406	.0092852	0	.1788618	539
Left rural	.1100383	.0002042	.0016842	0	.0362172	539
Productivity	451.0088	.8367511	1.404919	-3.751171	6.224155	539
Wage	110143.5	204.3478	77.39163	62.8971	447.6711	539
Gravity volume rural	52.34537	.0971157	.2547768	0	3.770528	539
Local volume rural	904.2745	1.677689	13.01002	0	237.0662	539
Printing house	8456	15.68831	20.912	0	143	539
Literate share rural	118.3467	.2195672	.1395503	.1004122	.692866	539
Observations	539					

	Worker strikes: Summary statistics					
	sum	mean	sd	min	max	count
Strikes urban	12.71634	.0288352	.0775	0	.8043487	441
Strikes part urban	4362.075	9.891326	34.74157	0	317.8284	441
Left urban	.1247171	.0002828	.0017815	0	.0293738	441
Productivity	422.2047	.9573804	1.439679	-3.751171	6.224155	441
Wage	88844.89	201.4623	76.81616	62.8971	447.6711	441
Gravity volume urban	182.2618	.4132921	.789534	0	8.508294	441
Local volume urban	902.3308	2.046102	16.35461	0	245.5653	441
Printing house	6040	13.69615	20.23079	0	143	441
Literate share urban	211.83	.4803401	.0948552	.3284435	.8030924	441
Observations	441					

4 Hypothesis and main results

4.1 Hypothesis

In my research, I am going to explore the correlation between labor leftist media (periodical print media) and popular unrest (peasant uprisings and worker strikes). There are several theoretically possible directions of the link between these variables. Positive direction: agitators, by distributing the leftist press, might intensify the emerging social conflict between workers (or peasants) and the state. Also, as it was already discussed, there might be a reverse causality. For example, agitators could spread the leftist press in gubernias with high levels of discontent in order to destabilize the situation even more. These two situations under consideration may be the reason for the positive relationship between the above variables. On the other hand, the link might be negative: if the left press adopted an extremely confrontational tone towards the tsarist regime, it may have alienated more moderate or cautious peasants and workers. In the case of reverse causality, the leftist parties could spread their media in the relatively peaceful gubernias to provide the initial impetus for the beginning of popular discontent. These options may explain the negative relationship between the variables. I hypothesize that the distribution of politically left-wing periodicals took place near the place of printing, since the movement of agitators over long distances leads to more costs: increased risk to be caught by the police, costs on train tickets, time costs, etc. Therefore, I assume that the correlation between the left press and popular unrest will be stronger the closer it is to the analyzed gubernia. Further, I will apply the gravity model method, which will discount the correlation between left-wing press and popular unrest by the distance between the analyzed region and the print region of the periodical.

4.2 Main models: Peasant revolts

My basic model for studying the correlation of the leftist press and peasant revolts is as follows:

$$Revolts\ rural_{it} = \alpha + \beta_1 \cdot Left\ rural_{it} + \beta_2 \cdot Productivity_{i,t-1} + \beta_3 \cdot Wage_{it} + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

$Revolts\ rural_{it}$ is the number of peasant revolts in gubernia i in year t divided by the rural population of that gubernia in year 1897 in thousands of people. $Left\ rural_{it}$ is the number of unique titles of left-wing periodicals in gubernia i in year t divided by the rural population of that gubernia in year 1897 in thousands. γ_i - is the gubernia fixed effect. δ_t - is the year fixed effect. D_i - is the dummy of gubernia i, so ζ_i is a linear time trend for gubernia i. I use fixed effects by gubernia to isolate variation in peasant revolts

associated with the geographic and historical characteristics of a particular province (e.g., soil quality and historical prevalence of serfdom (Finkel, Gehlbach, and Kofanov, 2017)[2], cultural characteristics, and national composition). Fixed effects for the year are used to isolate the effects of any global historical events that may affect popular discontent. Linear trends were included to estimate long-run trends in popular discontent by province. For example, due to population growth, the number of peasant revolts may have increased over the long-run, or growing level of urbanization might lead to more worker strikes. For normalization I use the rural population in a fixed 1897 year to avoid endogeneity, since the number of revolts and number of population can hypothetically directly influence each other.

$$\textit{Revolts rural}_{it} = \frac{\text{Revolts number}_{it}}{\text{Rural popul}_{i,1897}}$$

$$\textit{Left rural}_{it} = \frac{\text{Left outlet number}_{it}}{\text{Rural popul}_{i,1897}}$$

Also, I included control variables. The main metric that changes over time and needs to be controlled is people's standard of living. Of course, there are many other factors: morbidity rates, military conscription, price level, taxation level, etc. All of these metrics can affect the level of popular discontent, but it is hard to extract data on them. That is why I focus primarily on the economic well-being of citizens. In the case of peasants, this metric is crop yields, in the case of workers, wages. Here $\textit{Productivity}_{i,t-1}$ - is the level of agricultural productivity deviation from its trend. As it was mentioned above, this metric was constructed following the logic of the thesis by Gilyov[3]. At first I calculated the sum of poods of harvest divided by the sum of poods of seed inputs. For 1890-1899, I calculated the average level of this productivity for each of the gubernias. I considered it as a constant trend, which I subtracted from subsequent productivity indicators to calculate the deviation of yields from their normal historical values.

$$\text{Productivity trend}_i = \frac{\frac{\text{harvest}_{1890}}{\text{seeds}_{1890}} + \ldots + \frac{\text{harvest}_{1899}}{\text{seeds}_{1899}}}{10}, \quad i - \text{gubernia identifier}$$

$$\textit{Productivity}_{it} = \frac{\text{harvest}_{it}}{\text{seeds}_{it}} - \text{Productivity trend}_i$$

Following the ideas from the research of Waldinger (2023)[4], where it was discovered that the previous year crop failure was a significant determinant of peasant participation in the

[2]Finkel, E., S. Gehlbach, and D. Kofanov (2017). (good) land and freedom (for former serfs): Determinants of peasant unrest in european russia, march–october 1917. Slavic Review 76 (3), 710–721.

[3]Gilyov, I. (2021). "The Effect of Economic Shocks on Peasant Uprisings: Evidence from the Russian Empire." Bachelor's thesis, New Economic School & Higher School of Economics

[4]Waldinger, M. (2024). "let them eat cake": drought, peasant uprisings, and demand for institutional change in the french revolution. Journal of Economic Growth 29 (1), 41–77

French Revolution, I considered lag of the productivity as a control, since peasants eat their crops throughout the year. Hence, if the harvest turns out to be bad, it should not immediately affect popular discontent. They must first run out of food, which will probably happen after winter by next year. $Wage_{i,t}$ is the average wage of an urban worker in rubles. I include this indicator as a control, considering it as an opportunity cost for peasants who might prefer to go to work in the city at a sufficiently high wage rather than start revolting. In this and in all of the following specifications I use the clustering of errors by gubernias.

Revolts rural	(1) OLS	(2) OLS+Controls	(3) OLS	(4) OLS + Controls
Left rural	-0.0855	-0.0716	-0.0964	-0.0854
	(-1.20)	(-1.02)	(-1.21)	(-1.09)
$Productivity_{t-1}$		-0.000294		-0.000192
		(-1.11)		(-0.65)
Wage		-0.00000968		-0.00000875
		(-0.72)		(-0.44)
Constant	-0.00107*	-0.000310	0.107***	0.0487
	(-1.99)	(-0.12)	(9.95)	(0.35)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	539	539	539	539

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

It can be seen, that there is no statistically significant correlation between peasant revolts and left periodical press in specifications without linear gubernia trends (1,2) and with them (3,4).

In addition to the basic OLS regressions, I consider the following gravity model:

$$Revolts\ rural_{it} = \alpha + \beta_1 \cdot Gravity\ volume\ rural_{it} + \beta_2 \cdot Local\ volume\ rural_{it} +$$

$$+\beta_3 \cdot Productivity_{i,t-1} + \beta_4 \cdot Wage_{it} + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

My gravity model is organized as follows. In my work I consider 49 gubernias of the European part of the Russian Empire. I have identified the latitudes and longitudes for the centroids of each of these gubernias. Then for each gubernia in each year I determined the total circulation of the leftist press published in that gubernia. Thus, the variable *Gravity volume rural*$_{it}$ is the sum of the circulation of the leftist press issued in other gubernias divided by the distance between the centroids of the gubernias. Also, I normalized this metric to the rural population in thousands of people in 1897 for this gubernia.

$$\textit{Gravity volume rural}_{it} = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{49} \frac{\text{Total circulation}_{jt}}{\text{distance}_{ij}}}{\text{Rural popul}_{i,1897}}$$

The total circulation of leftist press issued directly in the considered gubernia is recorded in a separate variable *Local volume rural* to avoid division on zero distance. This metric is also divided by the rural population of this gubernia in thousands of people in 1897.

$$\textit{Local volume rural}_{it} = \frac{\text{Total circulation}_{it}}{\text{Rural popul}_{i,1897}}$$

It is important to note that circulation data are not known for all periodicals. Therefore, I made an approximation with similar outlets, grouping them by general type (newspapers, magazines, leaflets) and considered that in the missing data the circulations will be equal to the average value for their group.

Again, none of the variables considered were found to be statistically significant in the regressions without linear trends (1,2) and with them (3,4). The same results were obtained using the regressions in logarithms (Appendix, table 1). One possible explanation for the lack of correlation is that peasants were massively illiterate. They simply could not read newspapers, so the politically leftist press had no influence on their behavior or these press were not even distributed in villages, since agitators, thinking strategically, knew it was useless. I will examine this hypothesis in more detail when the correlation between left press and worker strikes is analyzed. Another possible reason for the lack of correlation is that I use the number of peasant revolts in these specifications. Revolts could have been both large and small, but I do not have data on the number of participants in the revolts and consider all of them to be equal entities. Thus, this limitation due to the lack of necessary data may introduce noise in the estimation of the real correlation between the leftist press and peasant revolts, leading to a lack of statistical significance.

Revolts rural	(1) Gravity	(2) Gravity+Controls	(3) Gravity	(4) Gravity + Controls
Gravity volume rural	-0.00123	-0.00117	-0.00164	-0.00164
	(-1.23)	(-1.27)	(-1.45)	(-1.52)
Local volume rural	-0.00000969	-0.00000762	-0.0000103	-0.00000848
	(-1.13)	(-0.86)	(-1.09)	(-0.84)
$Productivity_{t-1}$		-0.000276		-0.000170
		(-1.07)		(-0.59)
Wage		-0.0000101		-0.00000953
		(-0.75)		(-0.48)
Constant	-0.000972*	-0.0000444	0.0929***	0.0294
	(-1.72)	(-0.02)	(11.43)	(0.21)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	539	539	539	539

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4.3 Main models: Worker strikes

The basic model for worker strikes is quite similar to the case with peasants:

$$Strikes\ urban_{it} = \alpha + \beta_1 \cdot Left\ urban_{it} + \beta_2 \cdot Productivity_{i,t-1} + \beta_3 \cdot Wage_{it} + + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

$$Strikes\ urban_{it} = \frac{\text{Strikes number}_{it}}{\text{Urban popul}_{i,1897}}$$

$$Left\ urban_{it} = \frac{\text{Left outlet number}_{it}}{\text{Urban popul}_{i,1897}}$$

Here, the variable $Strikes\ urban_{it}$ is the number of strikes, divided by urban population of gubernia i in year 1897 in thousands of people. Left urban_{it} is the number of unique titles of left-wing periodicals in gubernia i in year t divided by the urban population of that gubernia in year 1897 in thousands. Again, I used fixed urban population for each gubernia to avoid endogeneity problem. It is important to note that the factory inspectorate did not report data on worker strikes from 1905 to 1908 due to their massive scale. Therefore, in this regression I consider data for 1901-1904 and 1909-1913. This specification, both without (1,2) and with (3,4) linear trends, showed no statistically significant correlations.

Strikes urban	(1) OLS	(2) OLS+Controls	(3) OLS	(4) OLS + Controls
Left urban	4.401	4.547	1.918	2.068
	(0.85)	(0.88)	(0.75)	(0.79)
$Productivity_{t-1}$		0.00383		0.00151
		(1.61)		(0.57)
Wage		0.0000982		0.000128
		(0.70)		(0.57)
Constant	0.0115	0.00765	-12.56***	-11.86***
	(0.58)	(0.19)	(-4.23)	(-4.66)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.01$, ** $p < 0.05$, *** $p < 0.01$

Further I estimate the gravity model, the same way I did for peasant revolts. However, here I normalized volumes of left press on the urban population.

$$\textit{Strikes urban}_{it} = \alpha + \beta_1 \cdot \textit{Gravity volume urban}_{it} + \beta_2 \cdot \textit{Local volume urban}_{it} +$$

$$+\beta_3 \cdot \textit{Productivity}_{i,t-1} + \beta_4 \cdot \textit{Wage}_{it} + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

$$\textit{Gravity volume urban}_{it} = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{49} \frac{\text{Total circulation}_{jt}}{\text{distance}_{ij}}}{\text{Urban popul}_{i,1897}}$$

$$\textit{Local volume urban}_{it} = \frac{\text{Total circulation}_{it}}{\text{Urban popul}_{i,1897}}$$

Strikes urban	(1) Gravity	(2) Gravity+Controls	(3) Gravity	(4) Gravity + Controls
Gravity volume urban	0.0146	0.0153	0.00623	0.00657
	(0.74)	(0.77)	(0.39)	(0.40)
Local volume urban	0.000986**	0.000978**	0.000535*	0.000535*
	(2.34)	(2.34)	(1.99)	(1.98)
$\textit{Productivity}_{i,t-1}$		0.00284		0.00108
		(1.32)		(0.37)
Wage		0.000133		0.000131
		(0.89)		(0.59)
Constant	0.0128	-0.00324	-12.29***	-11.36***
	(1.09)	(-0.08)	(-4.40)	(-3.36)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As it can be seen from the gravity model results, the correlation between *Local volume urban*$_{it}$ and *Strikes urban*$_{it}$ is positive and statistically significant at the 5% significance level without linear trends (1,2) and at the 10% level with linear trends (3,4). Economically speaking, the increase in *Local volume urban*$_{it}$ by 1 SD is associated with increase in number of strikes in gubernia by almost 4 for regressions (1,2) and by 2 for regressions (3,4), which are substantial numbers. Interestingly, the results obtained fit well into the hypothesis that for the left-wing press to have significant correlation with popular unrest, it is necessary that these people can read. Thus, the workers, being much more literate people than the peasants, had the opportunity to read newspapers, which is why we observe a statistically significant positive correlation here and none for peasants. I will test this hypothesis later.

For worker strikes, just as it was for the peasant revolts, there is a risk that the estimate obtained above is biased because in the previous regressions all strikes, both small and large, were considered identical entities. Also, in additional regressions in logarithms (Appendix, table 3) no statistically significant correlation was found for worker strikes. Unlike the peasant revolts, the data on the number of workers who participated in strikes is known. In order to make sure that the results obtained are robust, I made regressions with *Strikes part urban* $= \frac{\text{Strikes participants}_{it}}{\text{Urban popul}_{i,1897}}$ as the dependent variable.

$$\textit{Strikes part urban}_{it} = \alpha + \beta_1 \cdot \textit{Left urban}_{it} + \beta_2 \cdot \textit{Productivity}_{i,t-1} + \beta_3 \cdot \textit{Wage}_{it} + + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

Strikes part urban	(1) OLS	(2) OLS+Controls	(3) OLS	(4) OLS + controls
Left urban	1776.2	1859.7	826.9	929.6
	(0.98)	(1.03)	(0.89)	(1.00)
Productivity$_{t-1}$		1.038		0.218
		(1.19)		(0.19)
Wage		0.0594		0.0976
		(1.06)		(1.15)
Constant	-0.428	-8.828	-2357.3**	-1813.4*
	(-0.06)	(-0.54)	(-2.19)	(-1.95)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As in the specifications with the number of strikes, no statistically significant correlation was found. Let us turn to the gravity model.

$$Strikes\ part\ urban_{it} = \alpha + \beta_1 \cdot Gravity\ volume\ urban_{it} + \beta_2 \cdot Local\ volume\ urban_{it} +$$

$$+\beta_3 \cdot Productivity_{i,t-1} + \beta_4 \cdot Wage_{it} + \gamma_i + \delta_t + \sum_{i=1}^{49} \zeta_i \cdot D_i \cdot t$$

Strikes part urban	(1) Gravity	(2) Gravity+Controls	(3) Gravity	(4) Gravity + controls
Gravity volume urban	4.541	4.897	2.840	3.214
	(0.67)	(0.72)	(0.49)	(0.54)
Local volume urban	0.404**	0.404**	0.237**	0.238**
	(2.68)	(2.66)	(2.06)	(2.06)
$Productivity_{t-1}$		0.646		0.0187
		(0.77)		(0.02)
Wage		0.0702		0.0992
		(1.23)		(1.22)
Constant	0.640	-11.90	-2256.4*	-1538.1
	(0.17)	(-0.76)	(-1.80)	(-1.06)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	No	No	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The results of the gravity model for the number of strike participants also show positive and statistically significant correlation for *Local volume urban*, but now after adding linear trends to the regression, the coefficient at *Local volume urban* is still statistically significant at the 5% level. Economically this result is also significant. The increase in *Local volume urban* by 1 SD is associated with the increase in the number of strike participants by 1600 people without linear trends (1,2) and by 950 people with linear trends (3,4). It is also worth noting that the additional regressions in logarithms in this case also show a statistically significant positive correlation at the 5% level (Appendix,

table 3). Thus, implementing additional regressions for the number of strike participants gave greater robustness to the results obtained for worker strikes. The statistically and economically significant correlation is indeed observed in this case. However, why was it absent in the case of peasants? In what follows, I will test the literacy hypothesis I have outlined above.

5 Instrumental variable

Before addressing the issue of literacy, I try to consider the causal relationship between the left-wing press and popular unrest. For this, I'm using the instrumental variable method. As an instrumental variable, I use the number of printing houses in the gubernia in 1904, following the logic from master thesis of Andrey Kolyadenko[5]. Due to the First Russian Revolution, the censorship was relaxed in 1905 which led to a significant increase in leftist press. The variable proposed by Kolyadenko[6] represents the number of printing houses, which is taken to be zero until 1905, and then take as a constant value thereafter. This variable construction is motivated by the fact that it is very difficult to change printing capacity quickly and the figure for 1904 may well approximate the value for subsequent years. The value equal to 0 before 1905 since left parties could not use these printing capacities because of censorship. I think that this variable may be used as an instrument, because is should be correlated with leftist press, while at the same time the number of printing houses in each gubernia should be exogenous towards peasant revolts and strikes.

$$Printing\ house_{it} = \begin{cases} 0, & t < 1905 \\ n_i, & t \geq 1905 \end{cases} \quad , \quad n_i - \text{Printing houses in gubernia i}$$

I instrumented only basic OLS models for peasant revolts, worker strikes and worker strikes participants, since in gravity model there are two potentially endogenous variables, but I have only one instrument. First stage regressions for strikes and strikes participants are the same. As it can be seen, the first-stage F value is very low ($F < 10$), indicating low correlation between the instrument and the variable of interest. I used LIML estimator (Limited-information maximum likelihood), since it has been found to be more resistant to weak instruments problems. However, the problem with weak instrument still persists. I also did the calculation for regressions without linear trends, but the low F value problem remained (Appendix, table 4). Thus, it was not possible to resolve the issue of the endogeneity of the relationship between the left-wing press and popular unrest with the data I have. Therefore, I return to the correlation analysis between those metrics. In the next block of work, I will consider the hypothesis of literacy as the reason for the

[5]Kolyadenko, A. (2016). "Media, Printing Capacities and Unrest: a Case of Imperial Russia." Master thesis, New Economic School

[6]Kolyadenko (2016)

statistical significance of the correlation in the gravity model for workers and insignificance for peasants.

First stage	(1) OLS+Controls *Left rural*	(2) OLS+Controls *Left urban*
Printing house	-0.000016	0.0000339
	(-1.09)	(1.23)
Constant	0.2047499	1.00204***
	(0.64)	(2.65)
Controls	Yes	Yes
FE Year	Yes	Yes
FE Gubernia	Yes	Yes
Gubernia Trend	Yes	Yes
First stage F-value	1.19	1.5
Observations	539	441

z statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Second stage	(1) OLS+Controls *Revolts rural*	(2) OLS+Controls *Strikes urban*	(3) OLS+Controls *Strikes part urban*
Left rural	1.927972		
	(0.39)		
Left urban		-34.86105	-18856.72
		(-1.02)	(-1.08)
$Productivity_{t-1}$	-0.0002918	0.0003717	-0.3933197
	(-0.66)	(0.09)	(-0.2)
Wage	-0.000016	-0.0002093	-0.083133
	(-0.61)	(-0.55)	(-0.43)
Constant	-0.2679498	28.78858	19968.2
	(-0.14)	(0.71)	(0.96)
FE Year	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes
Gubernia Trend	Yes	Yes	Yes
First stage F-value	1.19	1.5	1.5
Observations	539	441	441

z statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6 Testing the literacy hypothesis

Based on the 1897 census data, I calculated the number of people who can read for each gubernia in rural and urban areas. In order to test the hypothesis that left-wing publications had an insignificant correlation with peasant revolts and statistically significant correlation with worker strikes because of different literacy levels, I added to the basic gravity model the interaction of the left-wing press variables (*Gravity volume*$_{it}$ and *Local volume*$_{it}$) with the share of the literate population in rural and urban areas, respectively.

As it can be seen, with the addition of the interaction on the share of the literate urban population, the statistical significance of *Local volume urban*$_{it}$ for worker strikes (2) and worker strikes participants (3) disappeared. It is important to note that in the case of worker strikes (2) the interaction of urban literacy level with the *Local volume* variable is positive and statistically significant at 10% level. This evidence supports the hypothesis of literacy as the channel that provided the significance of the relationship between the leftist press and popular unrest. In the case of worker strikes participants (3) this interaction itself is not statistically significant. However, the sign of the coefficient of the variable *Local volume urban* decreased but still remained positive. The sign of the interaction of this variable with the share of literate urban population is also positive. Despite the fact that the interaction turned out to be statistically insignificant, jointly the variables of left volume with their interactions on literacy are statistically significant at the 1% level (F = 7.16). This result also partially supports the hypothesis put forward about literacy rates. Although my specifications include fixed effects and linear trends, so the literacy rate variation across provinces and its growth are already implicitly accounted for in my models, for a more rigorous confirmation of this hypothesis it is necessary to obtain annual data on the literacy of the population in the Russian Empire. Unfortunately, no such data is available at this time.

The results obtained support the idea that leftist press did have a statistically significant positive correlation with workers' strikes because workers were literate people and could read the propaganda disseminated by the leftist parties. For peasants we do not observe such an significance, probably because they were mostly illiterate. Therefore, there might be two options: either left-wing press had no effect on peasants because of that or left parties thinking strategically didn't distribute left press among peasants, knowing it would have no effect on their behaviour.

	(1) Gravity *Revolts rural*	(2) Gravity *Strikes urban*	(3) Gravity *Strikes part urban*
Gravity volume rural	0.00111		
	(0.32)		
Local volume rural	-0.000116		
	(-0.77)		
*Gravity volume rural*Lit % rural*	-0.00660		
	(-1.28)		
*Local volume rural*Lit % rural*	0.000288		
	(0.77)		
$Productivity_{t-1}$	-0.000173	0.00132	0.0613
	(-0.61)	(0.47)	(0.05)
Wage	-0.00000859	0.000122	0.101
	(-0.42)	(0.53)	(1.17)
Gravity volume urban		-0.0553	-13.02
		(-0.85)	(-0.46)
Local volume urban		-0.00214	0.126
		(-1.59)	(0.17)
*Gravity volume urban*Lit % urban*		0.0891	25.81
		(0.76)	(0.53)
*Local volume urban*Lit % urban*		0.00441*	0.155
		(1.69)	(0.12)
Constant	0.0819	-12.79***	-2186.7
	(0.50)	(-3.90)	(-1.34)
FE Year	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes
Gubernia Trend	Yes	Yes	Yes
Observations	539	441	441

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

7 Conclusion

The goal of my work is to examine the relationship between the leftist press in early 20th century Tsarist Russia and popular unrest. In this paper, I found a statistically significant positive correlation between the number of worker strikes (as well as for strikes participants) and left-wing political periodicals. In the case of peasant revolts, no such correlation was observed. I hypothesized that the presence of the correlation in the case of workers and its absence in the case of peasants is due to the different literacy levels of these social groups. Analyzing this hypothesis by adding the interaction of the left press variables with the shares of literate population in rural and urban areas, it was found that statistical significance of correlation in the case of worker strikes disappears. Interaction with literacy rate in case of worker strike was also found to be positive and statistically significant. This fact partially supports the hypothesis. Workers were much more likely to be able to read than peasants, so the leftist press either had no effect on peasant revolts, or the leftist parties did not distribute the press in the villages.

This finding is an important historical result in the context of Tsarist Russia. Workers, being more literate people, were more easily influenced by printed propaganda and therefore were (or at least were seen by leftist parties) the main driving class of revolutionary-leftist movements. Peasants, on the other hand, were not exposed to propaganda to the same extent, due to their illiteracy. In addition, this fact is important for historical economics in general, as these results give us deeper understanding of semi-agrarian autocratic societies of the time (Portugal, Bulgaria, Serbia, Romania, etc.) and the political processes that took place in them. Also, finding a statistically significant correlation in the case of worker strikes is an important result within the field of media economic studies. As mentioned earlier, most studies in this area are conducted on data from democratic states, due to their greater availability. In turn, my research is conducted on data from authoritarian state. I found that, even with state censorship, there is some connection between opposition media and their audiences.

Of course, my paper does not cover the topic of leftist media in Tsarist Russia in its entirety. I see several possible ways to further develop this research. First of all, the resolution of causality problem is crucial for the understanding whether the opposition leftist media had the effect on the action of people or not. Secondly, the database I use has serious limitations: the number of worker strikes and their participants from 1907 to 1908 is unknown, since the factory inspectors did not keep records, and there is no decomposition of data at the uezd level. Also, there is no data on Siberian and Central Asian gubernias of the Russian Empire, the sampling is limited to the early 20th century, and literacy levels are known only for 1897. In addition, it might be interesting to analyze

other media resources, such as public speeches of agitators, possibly based on the data on political crimes. This line of analysis is of additional interest because in the case of public speeches, the literacy of the people the propaganda is trying to influence is not required.

8 Appendix

Table 1: Peasant revolts: logarithms

	(1)	(2)	(3)	(4)
	ln(*Revolts rural*)	ln(*Revolts rural*)	ln(*Revolts rural*)	ln(*Revolts rural*)
ln(*Left rural*)	0.122	0.132		
	(0.67)	(0.74)		
$Productivity_{t-1}$		-0.0178		-0.0186
		(-0.66)		(-0.70)
Wage		-0.00252		-0.00256
		(-1.25)		(-1.28)
ln(*Gravity volume rural*)			-0.0790	-0.0881
			(-0.86)	(-1.02)
ln(*Local volume rural*)			0.00742	0.00741
			(0.32)	(0.32)
Constant	56.72***	39.37**	55.70***	38.21**
	(5.45)	(2.11)	(4.68)	(2.15)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	Yes	Yes	Yes	Yes
Observations	539	539	539	539

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Worker strikes: logarithms

	(1)	(2)	(3)	(4)
	ln(*Strikes urban*)	ln(*Strikes urban*)	ln(*Strikes urban*)	ln(*Strikes urban*)
ln(*Left urban*)	0.149	0.170		
	(0.99)	(1.22)		
$Productivity_{t-1}$		0.0809		0.0830
		(1.23)		(1.27)
Wage		0.00417		0.00424
		(1.00)		(1.00)
ln(*Gravity volume rural*)			0.0175	0.0600
			(0.15)	(0.52)
ln(*Local volume rural*)			0.0488	0.0557
			(1.30)	(1.58)
Constant	-152.2***	-127.2***	-157.7***	-128.7***
	(-6.63)	(-3.82)	(-8.24)	(-3.90)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	Yes	Yes	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Worker strikes participants: logarithms

	(1)	(2)	(3)	(4)
	ln(*Strikes part urban*)	ln(*Strikes part urban*)	ln(*Strikes part urban*)	ln(*Strikes part urban*)
ln(*Left urban*)	0.629	0.677		
	(1.49)	(1.68)		
$Productivity_{t-1}$		0.172		0.169
		(1.07)		(1.08)
Wage		0.00994		0.00967
		(1.08)		(1.05)
ln(*Gravity volume rural*)			-0.0784	0.0122
			(-0.28)	(0.04)
ln(*Local volume rural*)			0.153**	0.168**
			(2.23)	(2.64)
Constant	-91.07	-31.12	-97.81*	-31.83
	(-1.41)	(-0.35)	(-2.19)	(-0.44)
FE Year	Yes	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes	Yes
Gubernia Trend	Yes	Yes	Yes	Yes
Observations	441	441	441	441

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: IV regressions without linear trends

First stage	(1) OLS+Controls *Left rural*	(2) OLS+Controls *Left urban*
Printing house	-0.00000451	0.00000734
	(-0.72)	(0.93)
Constant	-0.0001211	0.0048749***
	(-0.14)	(4.25)
Controls	Yes	Yes
FE Year	Yes	Yes
FE Gubernia	Yes	Yes
Gubernia Trend	No	No
First stage F-value	0.52	0.86
Observations	539	441

z statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Second stage	(1) OLS+Controls *Revolts rural*	(2) OLS+Controls *Strikes urban*	(3) OLS+Controls *Strikes part urban*
Left rural	-3.29409		
	(-0.37)		
Left urban		-0.1988905	14822.23
		(-0.01)	(0.69)
$Productivity_{t-1}$	-0.000123	0.003834	1.030958
	(-0.20)	(1.27)	(0.61)
Wage	-0.00000361	0.0000634	0.1542245
	(-0.15)	(0.2)	(0.89)
Constant	-0.0004857	0.0303049	-70.6972
	(-0.09)	(0.16)	(-0.67)
FE Year	Yes	Yes	Yes
FE Gubernia	Yes	Yes	Yes
Gubernia Trend	No	No	No
First stage F-value	0.52	0.86	0.86
Observations	539	441	441

z statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

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